

Name : _____

ID Number: _____

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BITS Pilani, Hyderabad Campus
BITS F464 Machine Learning
Mid Semester Examination (Regular), First Semester (2023-24)

Date: October 9th, 2023

Time: 11.30 am - 13.00 pm

Maximum Marks: 60

Instructions:

1. Answer all the questions.
2. Provide clear and concise answers. Answers should be limited to the space provided else will not be considered.
3. Make assumptions, wherever necessary and state the same clearly.
4. Use of calculators is allowed.

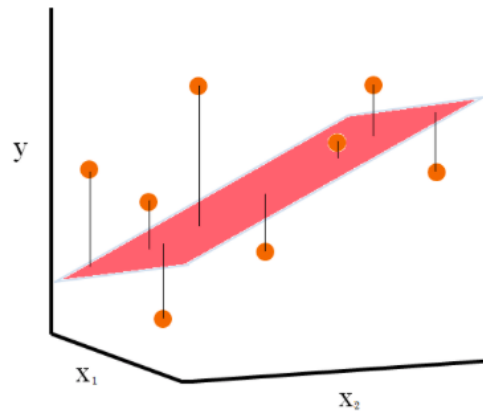
NOTE: To be update only by the examiners

<i>Q1 a (3 marks)</i>	<i>Q1 b (3 marks)</i>	<i>Q1 c (6 marks)</i>	<i>Q1 d (6 marks)</i>
<i>Q2 (14 marks)</i>	<i>Q3 (14 marks)</i>	<i>Q4 (14 marks)</i>	<i>Total (60 marks)</i>

Q1 Short answer type questions. Provide answers in 2-3 sentences.

a) Whether learning rate should be slow or fast in gradient descent algorithm? What could happen in either of the two cases? [3 marks]

- b) Refer to the figure below and provide an explanation about the type of model represented? [3 marks]



- c) How can high bias and variance be detected in a given machine learning model? [6 marks]

- d) How can a complete second degree polynomial model in two variables x_1 and x_2 be represented? Give an equation and explain the different terms involved in the expression. [6 marks]

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Q2 a) What is the correspondence between the Naive Bayes and Logistic regression algorithms? Under what assumptions are the two algorithms equivalent to one another. [6 marks]

b) Show the derivation depicting the relation between the two for a binary classification problem.

[8 marks]

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Q3 a) Given the following raw values: $[3.2, -5.7, 0.6]$. Show the results of sigmoid and softmax calculations for all three output values. Using the results, highlight the difference between sigmoid and softmax functions. [8 marks]

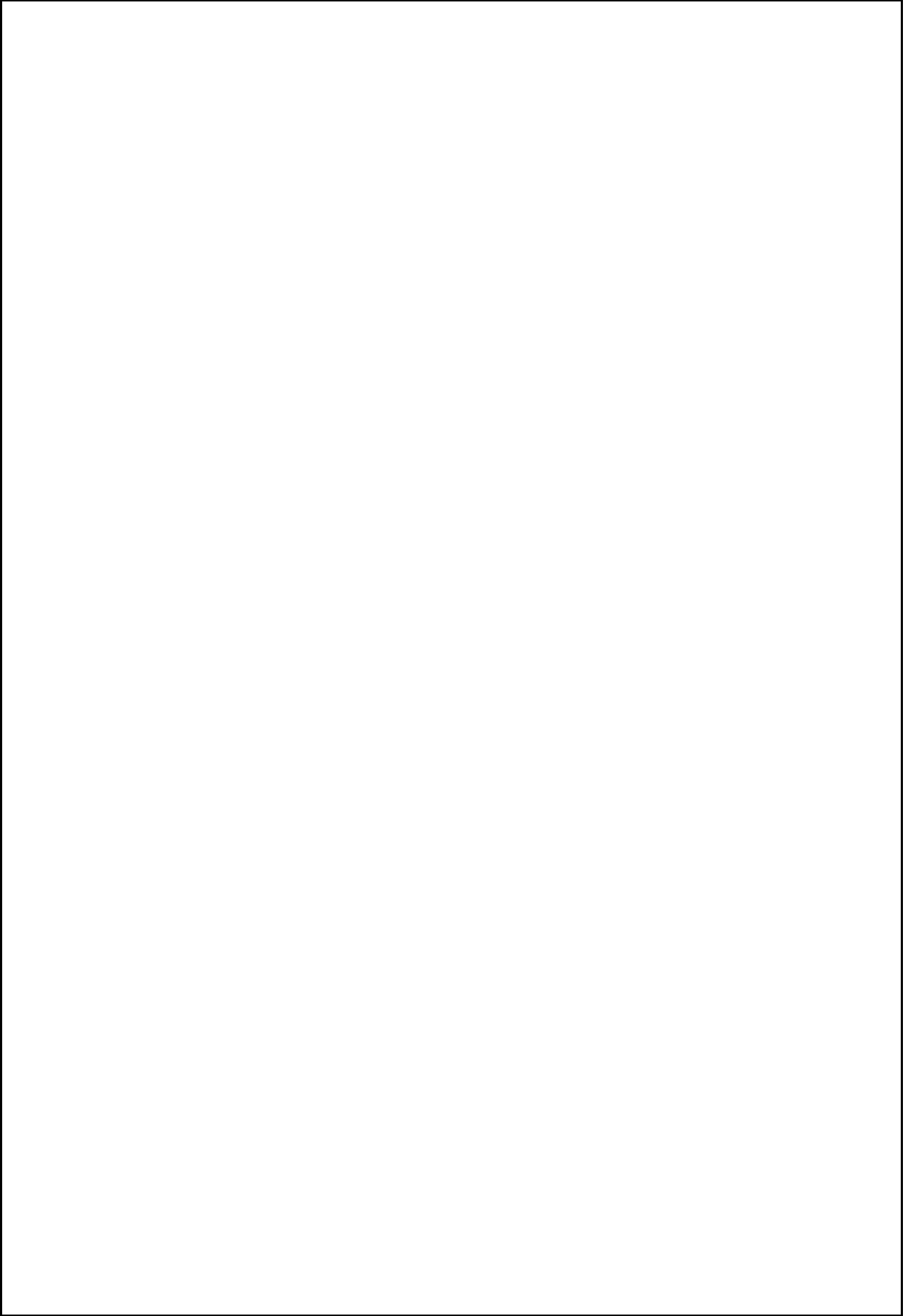
Q3 b) Show that the softmax function reduces to sigmoid in case of binary classification with number of classes $k=2$. [6 marks]

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Q4. Compute the Linear Discriminant projection for the following two dimensional dataset. Samples for class ω_1 : $X_1 = (x_1, x_2) = \{(4,2), (2,4), (2,3), (3,6), (4,4)\}$ and samples for class ω_2 : $X_2 = (x_1, x_2) = \{(9,10), (6,8), (9,5), (8,7), (10,8)\}$ [14 marks]



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Some useful formulae

$$\begin{aligned} p(\mathcal{C}_1|\mathbf{x}) &= \frac{p(\mathbf{x}|\mathcal{C}_1)p(\mathcal{C}_1)}{p(\mathbf{x}|\mathcal{C}_1)p(\mathcal{C}_1) + p(\mathbf{x}|\mathcal{C}_2)p(\mathcal{C}_2)} & p(\mathcal{C}_k|\mathbf{x}) &= \frac{p(\mathbf{x}|\mathcal{C}_k)p(\mathcal{C}_k)}{\sum_j p(\mathbf{x}|\mathcal{C}_j)p(\mathcal{C}_j)} \\ &= \frac{1}{1 + \exp(-a)} = \sigma(a) & &= \frac{\exp(a_k)}{\sum_j \exp(a_j)} \end{aligned}$$

$$\mathcal{N}(x|\mu, \sigma^2) = \frac{1}{(2\pi\sigma^2)^{1/2}} \exp\left\{-\frac{1}{2\sigma^2}(x - \mu)^2\right\}$$

Note - USE ONLY FOR ROUGH WORK - This page will not be marked.